AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

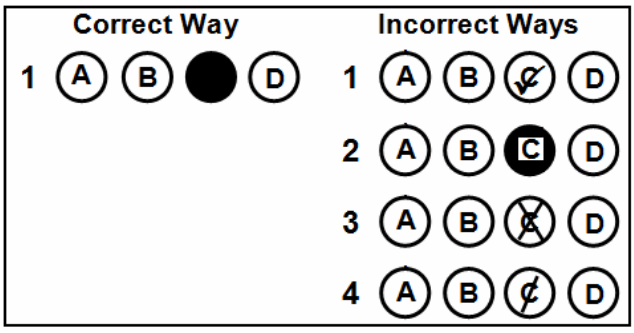
MAY 2013

Physics Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

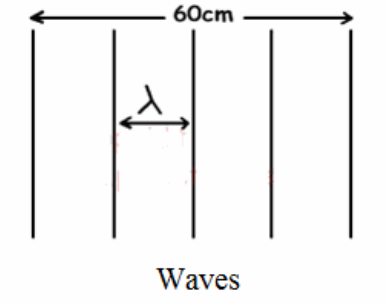
5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. Which of the following depicts the appropriate change in wave length and speed of water waves, when they pass from deep to shallow water?

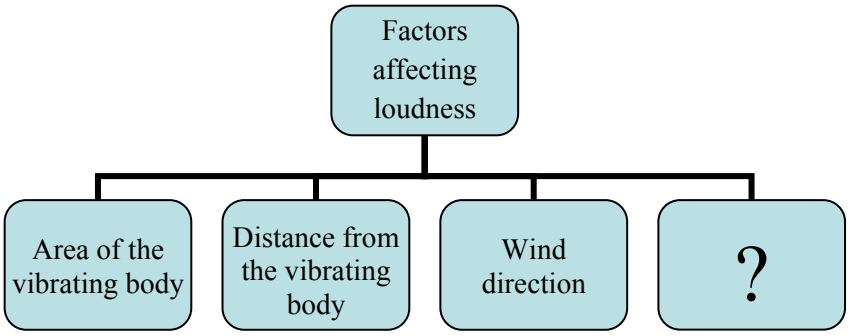
	Wavelength	Speed
A	Decreases	Decreases
B	No change	Decreases
C	Increases	No change
D	Increases	Increases

2. In the given figure the frequency of the wave source is 3 Hz. The velocity of the wave will be

A. 0.5 cm/s
B. 12 cm/s
C. 20 cm/s
D. 45 cm/s

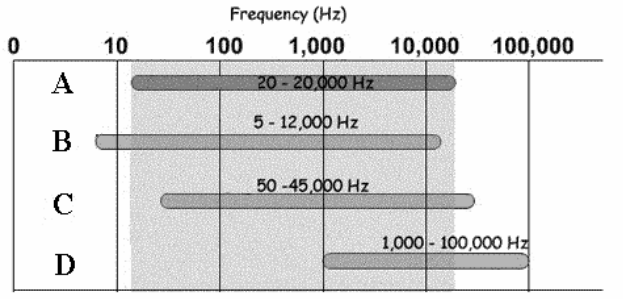


3. Which of the following is the fourth factor upon which the loudness of sound depends?



A. Amplitude
B. Frequency
C. Time period
D. Wave length

4. Which of the following options represents the range of frequencies of sound that can be heard by a normal human ear?



5. When an object is placed between the principal focus and the centre of curvature of a convex lens, then the image will be

A. real.
B. erect.
C. virtual.
D. small in size.

6. The focal length of a spherical mirror is equal to

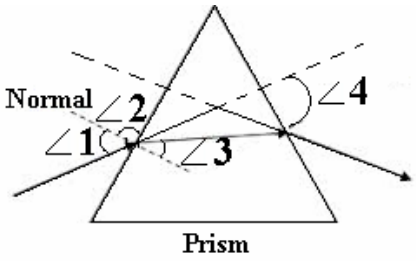
A. the radius of curvature.
B. half of the radius of curvature.
C. one third of the radius of curvature.
D. one fourth of the radius of curvature.

7. If the real depth of a swimming pool is 2 m and the refractive index of water is 1.33, then the apparent depth of the pool will be

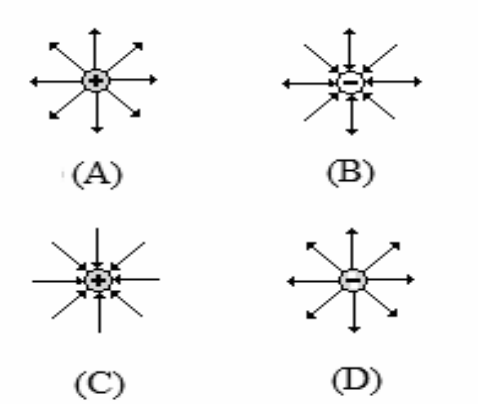
A. 1.0 m
B. 1.5 m
C. 2.5 m
D. 3.0 m

8. In the given diagram, a light ray is passing through a prism. Which of the following angles represents the angle of deviation?

A. $\angle 1$
B. $\angle 2$
C. $\angle 3$
D. $\angle 4$



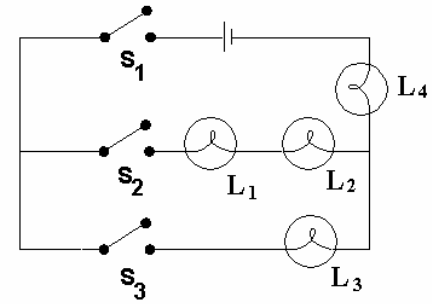
9. Which of the following is the correct representation of the electric field lines?



10. When two capacitors of the capacitance $4\,\mu F$ and $6\,\mu F$ are connected in series, then the equivalent capacitance will be

A. $2.4\,\mu F$
B. $3.4\,\mu F$
C. $4.4\,\mu F$
D. $5.4\,\mu F$

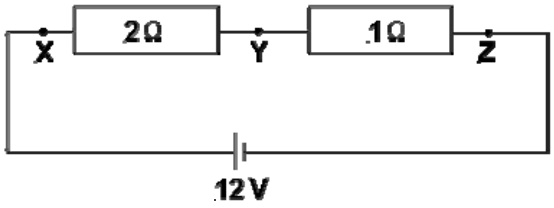
11. In the given circuit diagram, four identical lamps L_1, L_2, L_3 and L_4 are connected and controlled by switches S_1, S_2 and S_3.



Which of the following lamps will light when the switches S_1 and S_2 are closed?

A. Lamp L_1 and L_2
B. Lamp L_2 and L_3
C. Lamp L_2, L_3 and L_4
D. Lamp L_1, L_2 and L_4

12. A circuit diagram is shown below.



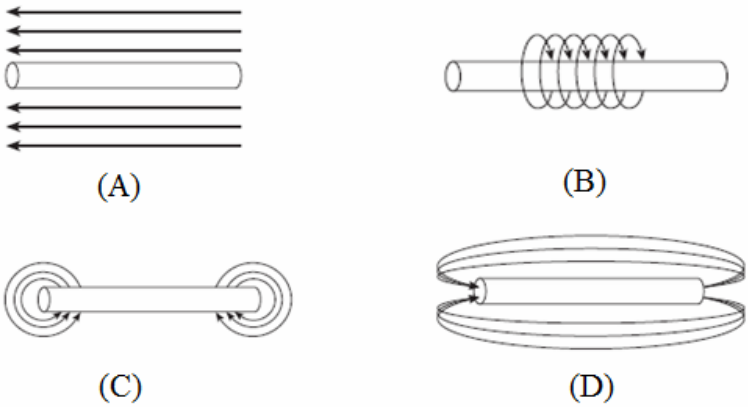
The potential difference between point X and point Y will be

A. 3 volt
B. 6 volt
C. 8 volt
D. 12 volt

13. The given diagram shows that current is flowing through a wire.



Which of the following options represents the magnetic field produced by the current in the wire?



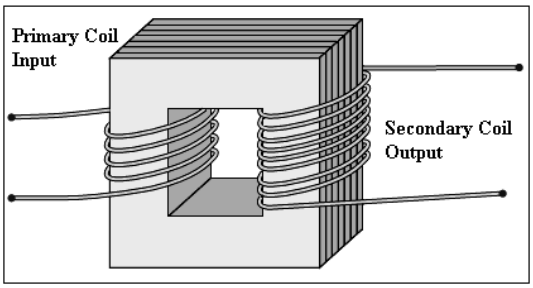
14. Which of the following transformers is used for decreasing output voltage?

Transformer	Number of Turns in Primary Coil	Number of Turns in Secondary Coil
A	20	30
B	40	80
C	150	90
D	200	400

15. Which of the following devices is used to increase or decrease Alternating Current?

A. Ammeter
B. Generator
C. Ohmmeter
D. Transformer

16. The diagram below shows a transformer.



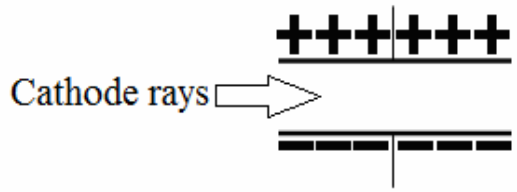
Which of the following statements is correct about the transformer?

A. It is used to generate electricity.
B. It is used in a direct current circuit.
C. The input voltage is less than the output voltage.
D. The input voltage is greater than the output voltage.

17. In thermionic emission, when a filament is heated a beam of tiny particles emits from it. These particles are known as

A. protons.
B. neutrons.
C. electrons.
D. alpha particles.

18. A beam of cathode rays passes through an electric field between two parallel plates.



Which of the following changes will take place after the beam has passed through the plates?

A. It will bounce back.
B. It will pass un-deflected.
C. It will bend towards the positive plate.
D. It will bend towards the negative plate.

19. Light is confined within the core of a simple optical fibre by

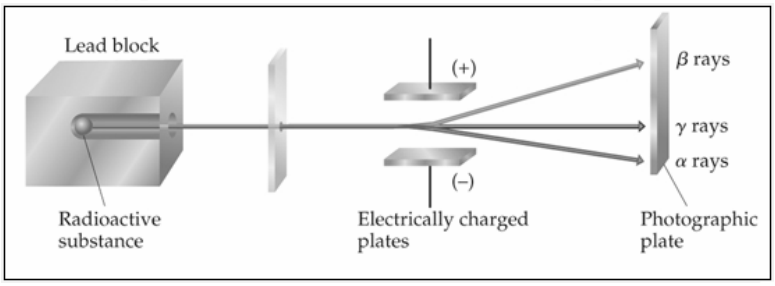
    A. reflection.
    B. refraction.
    C. diffraction.
    D. total internal reflection.

20. Which of the following electronic devices stores the largest amount of information?

    A. Hard disc
    B. Flash drive
    C. Compact disc
    D. Audio cassette

21. In the given diagram of a radioactive decay, β-particles are deflected more than α-particles because they have

    A. small mass.
    B. high velocity.
    C. high kinetic energy.
    D. more penetrating power.



22. The given equation shows the disintegration of Ra^{226} into Rn^{222} when a radioactive emission occurs. The radioactive emission is represented by X. What do you infer about X?

$$^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{A}_{Z}X$$

    A. A beta particle is emitted.
    B. An alpha particle is emitted.
    C. Only gamma radiation is emitted.
    D. 4 protons and 2 neutrons are emitted from the nucleus.

23. Which of the following charges is carried by a beta particle?

    A. Single positive
    B. Single negative
    C. Double positive
    D. Double negative

24. Which of the following is NOT affected by electric and magnetic fields?

    A. Alpha particles
    B. Beta particles
    C. Cathode rays
    D. Gamma rays

25. If half life of an element is 24 days, then how much mass of its 0.64 gm will remain unchanged after 48 days?

A. 0.08 gm
B. 0.16 gm
C. 0.32 gm
D. 0.64 gm

END OF PAPER